Designing a Universal Information Language

Oscar Bender-Stone

March 26, 2026

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Agenda

Introduction

Language Overview

Syntax

Metatheory

Conclusion

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Background

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- Information Management: studies how to **store** and **organize** data

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    - Data ⇒ Information ⇒ Wisdom
  - **Buckland:** Based on usage[2]
    - Can be a *process*, or *knowledge*
    - Most useful as *thing*
  - **Floridi:** Information is **truthful**[3]

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Problems

- How to *actually* manage information?
  - **Information base:** a database of information
- **Depth:** domain specific data
  - Example: Amazon (product suggestions)[1]

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  - **Information base:** a database of information
- **Depth:** domain specific data
  - Example: Amazon (product suggestions)[1]
- **Breadth:** different kinds of data
  - Example: Wikipedia[2]
- **Standardization:** ambiguous interpretations
  - Related issue: being **interoperable**
  - **Interoperable:** a portable data format
  - Example: US health records[3]

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Existing Solutions

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Existing Solutions

- **Resource Description Language (RDF):**[1]
  - ▸ Used in Web Ontology Language **(OWL)**
  - \+ Built to be **interoperable**
  - — No built-in logic engine
- **Labeled Property Graphs (LPGs):**[2]
  - ▸ Stores **properties** (node/edge metadata)
  - \+ Efficient queries
  - — No logic engine. Hacky tooling/scripts
- **Cyc:**[3]
  - ▸ Largest **knowledge base** to date
  - \+ Sophisticated logic engine
  - — Hard-coded, proprietary rules

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Proposed Solution

- **Welkin**: an information language
  - Pronounced: “VEL-KIN”
  - Old German word, meaning “cloud”, “sky”[1]
- **Goals:**

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- **Welkin**: an information language
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- **Goals:**
  - **Universality:** be completely expressive (**done**)
    - *Any* computer program
    - *Any* "reliable" proof
  - **Scalability:** work at any size (**WIP**)
    - Be efficient
    - Support: future optimizations
    - Describe improvements *in* Welkin
  - **Standardization:** be completely *unambiguous* (**WIP**)

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  - **Standardization:** be completely *unambiguous* (**WIP**)
    - Have a formal specification

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General Requirements

1. Express any computer program

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General Requirements

1. Express any computer program
   - Need effective operations
   - Established notion: **Turing machines**[1]
2. Enable user-defined *meaning*

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- Achieve both via **representations**

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- Track *faithfulness* of representations via **truth**

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- Can be broken down, combined
  - Written with braces $\{a, b, \ldots, z\}$

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- Representations are units
- Each unit has its own **context**

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Information

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Information

- Representations

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- Representations ~ **Truth**

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  - More details later! (Metatheory)

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Rules: Primary

ID	Name	Rule
R1	Internal Transitivity	$a - c \to b$ and $b - c \to d$ imply $a - c \to d$
R2	Pair Congruence	If $a - c \to b$, then $\{a, d\} - c \to \{b, d\}$
R3	Sign Congruence	If $a - c \to b$, then $\{a - g \to d\} - c \to \{b - g \to d\}$
R4	Context Congruence	If $a - c \to b$, then $\{d - a \to g\} - c \to \{d - b \to g\}$
R5	Referent Congruence	If $a - c \to b$, then $\{d - g \to a\} - c \to \{d - g \to b\}$
R6	Contextual Lifting	$a - c \to b$ and $d - a \to g$ imply $\{d - b \to g\} \preccurlyeq c$
R7	Refinement	$\{a - c \to b, a - c \to d\} \longrightarrow \{a - c \to b\}$

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Rules: Empty Unit

ID	Name	Rule
R8	Empty Set	$\{g, \{\}\} \leftarrow c \rightarrow g$
R9	Sign Absorption	$\{\{\} - a \rightarrow b\} - c \rightarrow \{\}$
R10	Context Absorption	$\{a - \{\} \rightarrow b\} - c \rightarrow \{\}$
R11	Referent Absorption	$\{a - b \rightarrow \{\}\} - c \rightarrow \{\}$

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Rules: Lattice

ID	Name	Rule
R12	Handle Equality	If $h_1 = h_2$, then $h_1 \leftarrow c \rightarrow h_2$
R13	Derivation	$\{a - g \rightarrow a\} \leftarrow c \rightarrow \{\{g, a\} \longleftrightarrow g\}$
R14	Idempotency	$\{a, a\} \leftarrow c \rightarrow a$
R15	Associativity	$\{a, \{b, c\}\} \leftarrow c \rightarrow \{\{a, b\}, c\}$
R16	Commutativity	$\{a, b\} \leftarrow c \rightarrow \{b, a\}$

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Turing Completeness

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- Part of Universality

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  - $\langle\varphi_a\rangle$:

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  - $\langle \varphi_a \rangle$: encodes φ_a as a string
- Later:

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- Later: completeness for **proofs**!

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Encoding

- Characters written in `type-writer font`[1]
- Define US-ASCII: Printable (codes: 32-126) + `EOF` (code: 127)
- **Token:** collection of contiguous characters
  - Simplifies grammar

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[1]Font used: Intel One Mono (https://github.com/intel/intel-one-mono), licensed under OFL 1.1 (https://github.com/intel/intel-one-mono/blob/main/OFL.txt).

US-ASCII

Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph
32	20	Space	46	2E	.	60	3C	<	74	4A	J	88	58	X	102	66	f	116	74	t
33	21	!	47	2F	/	61	3D	=	75	4B	K	89	59	Y	103	67	g	117	75	u
34	22	"	48	30	0	62	3E	>	76	4C	L	90	5A	Z	104	68	h	118	76	v
35	23	#	49	31	1	63	3F	?	77	4D	M	91	5B	[	105	69	i	119	77	w
36	24	$	50	32	2	64	40	@	78	4E	N	92	5C	\	106	6A	j	120	78	x
37	25	%	51	33	3	65	41	A	79	4F	O	93	5D	]	107	6B	k	121	79	y
38	26	&	52	34	4	66	42	B	80	50	P	94	5E	^	108	6C	l	122	7A	z
39	27	'	53	35	5	67	43	C	81	51	Q	95	5F	_	109	6D	m	123	7B	{
40	28	(	54	36	6	68	44	D	82	52	R	96	60	`	110	6E	n	124	7C	|
41	29	)	55	37	7	69	45	E	83	53	S	97	61	a	111	6F	o	125	7D	}
42	2A	*	56	38	8	70	46	F	84	54	T	98	62	b	112	70	p	126	7E	~
43	2B	+	57	39	9	71	47	G	85	55	U	99	63	c	113	71	q			
44	2C	,	58	3A	:	72	48	H	86	56	V	100	64	d	114	72	r			
45	2D	-	59	3B	;	73	49	I	87	57	W	101	65	e	115	73	s			

Table 1: Printable US-ASCII codes and glyphs.

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Character Classes

Set Name	Characters
PRINTABLE	Listed in Table 1
WHITESPACE	\t | \r | Space
DELIMITER	{ | } | ' | " | . | ,
RESERVED	DELIMITER | * | @
SQ_CHAR	Any PRINTABLE except ' and \
DQ_CHAR	Any PRINTABLE except " and \
ID_CHAR	Any PRINTABLE not in RESERVED or WHITESPACE

Table 2: Key character classes.

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Building Blocks

- **Invertible Syntax Descriptions:** pairs of functions:[1]

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  - Identity: when parser succeeds, is "reversible" with printer.
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The Welkin Grammar

```
start := units - lexeme -> EOF,
units := "" | {unit - lexeme -> *{"",
"," - lexeme -> units}},
unit := node - lexeme -> *{
  ":=" - lexeme -> choices,
  *{"", arc - lexeme -> chain}
},
choices := *{"", "|"} - lexeme ->
choices_list,
choices_list := *{"",
  chain - lexeme -> *{"", "|" - lexeme
-> choice_list}
},
chain := *{"", "*"} - seq -> path -
lexeme -> *{"", arc - lexeme ->
chain},
arc := right_arc | other_arc,
right_arc := "-" - seq -> node - seq -
> "->",
other_arc := "<-" - seq -> node - seq
-> *{"-", "->"},
node := *{"", "*"} - seq -> path,
path := *{"", modifiers} - seq ->
trailer - seq -> *{"", graph},
modifiers := *{
  "@",
  "~" - seq -> *{"", "@"}
},
trailer := segment - seq -> *{"", "."
- seq -> trailer},
segment := HANDLE - lexeme -> *{"",
graph},
graph :=  "{" - lexeme -> units -
lexeme -> "}",
HANDLE := ID | STRING,
STRING := SQ_STRING | DQ_STRING,
SQ_STRING := "'" - seq -> *{SQ_CHAR,
"\'"} - seq_many_till -> "'",
DQ_STRING := '"' - seq -> *{DQ_CHAR,
'\"'} - seq_many_till -> '"',
ID := ID_CHAR - seq_many_till ->
*{RESERVED, WHITESPACE}
```

Listing 1: Full Welkin grammar.

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Outline of Unambiguity

- Simple grammar class: LL(1)

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[2]Edelmann et al. (2020). *Zippy LL(1) parsing with derivatives*. PLDI 2020.

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Metatheory

Conclusion

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Primer

- Terminology
  - **RE:** accepted by some Turing machine
    - Might not halt!
  - **Formal system:**
    - A decidable set of **formulas**

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  - **First Order Logic:**
    - Propositional logic (and, or, not)
    - Quantifiers (for all, exists)
  - **Peano Arithmetic** (PA): natural numbers + induction
  - **Consistency:** no proofs contain falsehood ($\perp$)

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Formal System Limits

- Many theorems on limit:s

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Artëmov's Selector Proofs

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Serial-Soundness

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Serial-Soundness

- Extend Artëmov to **serial-soundness:**
  - Secret: *same* technique!
  - Use **partial truh predicates**[1]
    - Approximates truth up to n derviations
- **Serial soundness:** a first-order RE theory T with a selector s.t.:
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- **Self-verifying**: *in* PA, T can prove its own serial-soundness
  - PA $\vdash_T \varphi$ if $T \vdash \varphi$ and T self-verifying

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Reliable Proofs

- How to generalize to *any* theory?

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  - p_φ: S-proof of φ

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  - HUGE leap; how to generalize?
  - Answer: comprehension!

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Crash Course: Recursive Ordinals

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  - Important: this is recursive if T is RE *and* sound

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Crash Course: Arithmetic Hierarchy

- Measures complexity of PA formulas
- Need to construct comprehension
  - Use Simpson's comprehension axiom:

$$\exists X.\forall n.(n \in X \Leftrightarrow \Phi(n))$$

  - Φ: very complex formula (Δ^1_1)

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Proof Completeness (con't)

- Let:
  - λ be limit of recursive $\beta_1, \beta_2, \ldots$
  - Each T_{β_i} self-verifying
  - $T' = \bigcup_{\beta<\lambda} T_\beta$
  - $\mathrm{Comp}_\lambda(\Delta^1_1) = \exists X_{\varphi,\lambda}.\forall n.(n \in X_{\varphi,\lambda} \Leftrightarrow \mathrm{PA} \vdash_{T'} \varphi(n))$
- Our construction: T_λ
  - ✓ Eventually reaches limit
  - ✓ Self-verifying
  - ✓ **Constructive**
- TL;DR: Artëmov + Simpson + Reliability = Proofs 💯

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Comparisons to Existing Solutions

- **Resource Description Framework:**
  - Similar ideas/terms about representations
  - Welkin:
    - Higher order terms
    - Better ogical reasoning
- **Labeled Property Graphs:**
  - Simple core idea + expressive
  - Welkin:
    - Formal definitions and grammar
    - Logical engine
- **Cyc:**
  - Similarly depend on contexts
  - Welkin:
    - Flexible data format

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- Open specification

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Future Work

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- Write the Standard

DRAFT

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- Create a Fast Implementation

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Thank you!

Questions?

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